Warmup/Attendance: [Week 6: Friday 10/2/2026]

Full Name: | SID:

Work with the people around you to try to solve these problems, which we'll discuss together once lecture begins. These are not graded for correctness, but you will turn this sheet in for attendance at the end of class.

1. Tossing a fair coin 3 times and counting the number of heads

Let X be the number of heads in **three** tosses of a fair coin. What is the probability mass function $f(x) = P(X = x)$ of X? What is this distribution called? What values will x take?

2. Tossing a biased coin 3 times and counting the number of heads

What about if X is the number of heads in 3 tosses of a biased coin, where the chance of heads is $\frac{2}{3}$? Now what is $f(x) = P(X = x)$?

3. Tossing a fair coin until it lands heads

Now suppose we toss a fair coin *until* the *first* time it lands heads, and let X be the number of **tosses** (including the last one, which is the first time the coin lands heads). What is the probability mass function of X? Is it binomial?

4. Dealing cards and counting hearts

Finally, let's consider a deck of cards, and we are interested in the *number* of hearts dealt in a hand of five. Call this number X. What is the pmf of X?